

Junkai Dong

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EDUCATION

Doctor of Philosophy in Physics Aug 2021 – May 2026

Harvard University, Cambridge, MA

Advisor: Ashvin Vishwanath

Thesis: *Two Silhouettes of Strongly Correlated Topological Phases: Fractionalization and Crystallization*

Master of Arts in Physics

Aug 2021 – May 2024

Harvard University, Cambridge, MA

Bachelor of Arts in Physics and Mathematics, *summa cum laude*

Aug 2017 – May 2021

Cornell University, Ithaca, NY

Thesis: *Averaging over deformed WZW models*

RESEARCH EXPERIENCE

- **Author of 15 peer-reviewed publications and 7 preprints**, with 1 in *Proc. Natl. Acad. Sci. USA*, 1 in *Phys. Rev. X*, and 6 in *Phys. Rev. Lett.*
- **Presented 14 invited talks** at scientific conferences and seminars.
- **Submitted a total of 53 referee reports** for *Nature*, *Phys. Rev. Lett.*, *Phys. Rev. X* and other journals.

RESEARCH ACHIEVEMENTS

- Proposed the concept of anomalous Hall crystals to explain experimental phenomenology in rhombohedral graphene samples. Paper is included in *Phys. Rev. Lett. Collection of the Year 2024*.
- Predicted the existence of zero-field composite Fermi liquids in twisted MoTe_2 , which was subsequently verified in experiments.
- Discovered a previously unknown kineo-elastic term present in elastic theory.

HONORS AND AWARDS

- KITP Graduate Fellow, Kavli Institute for Theoretical Physics, University of California Santa Barbara, 2024
- Gertrude and Maurice Goldhaber Prize, Harvard University, 2024
- Certificate of Special Recognition, Harvard Undergraduate Association, 2023
- Purcell Fellowship, Harvard University, 2021
- Donald R. Yennie Prize in Physics, Cornell University, 2021
- Bethe Thesis Prize in Physics, Cornell University, 2021

SKILLS

Languages Chinese (native), English (bilingual)

Programming Languages Julia, Python, C++

Software bash, slurm, OpenMP, MPI, MATHEMATICA, L^AT_EX

MANUSCRIPTS

First or Co-First Author

22. J. Dong^{*}, T. Soejima^{*}, D. E. Parker, and A. Vishwanath, “Crystals Caught Doping: Metallic Wigner Crystals in Rhombohedral Graphene”, *arXiv:2604.00114*.
21. J. Dong^{*}, O. E. Sommer^{*}, T. Soejima, D. E. Parker, and A. Vishwanath, “Phonons in Electron Crystals with Berry Curvature”, *Proc. Natl. Acad. Sci. USA* **122**(38), e2515532122 (2025). *arXiv:2503.16390*.
20. T. Soejima^{*}, J. Dong^{*}, A. Vishwanath, and D. E. Parker, “ λ -Jellium Model for the Anomalous Hall Crystal”, *Phys. Rev. Lett.* **135**, 186505 (2025). *arXiv:2503.12704*.
19. T. Soejima^{*}, J. Dong^{*}, T. Wang, T. Wang, M. P. Zaletel, A. Vishwanath, and D. E. Parker, “Anomalous Hall Crystals in Rhombohedral Multilayer Graphene II: General Mechanism and a Minimal Model”, *Phys. Rev. B* **110**, 205124 (2024). *arXiv:2403.05522*. (Editor’s Suggestion)
18. J. Dong^{*}, T. Wang^{*}, T. Wang^{*}, T. Soejima, M. P. Zaletel, A. Vishwanath, and D. E. Parker, “Anomalous Hall Crystals in Rhombohedral Multilayer Graphene I: Interaction-Driven Chern Bands and Fractional Quantum Hall States at Zero Magnetic Field”, *Phys. Rev. Lett.* **133**, 206503 (2024). *arXiv:2311.05568*. (Featured in PRL’s Collection of the Year 2024, Journal Club for Condensed Matter Physics, Nature News and Physics World.)
17. J. Dong, J. Wang, P. J. Ledwith, A. Vishwanath, and D. E. Parker, “Composite Fermi Liquid at Zero Magnetic Field in Twisted MoTe₂”, *Phys. Rev. Lett.* **131**, 136502 (2023). *arXiv:2306.01719*. (Editor’s Suggestion, Featured in *APS Physics*)
16. J. Dong, P. J. Ledwith, E. Khalaf, J. Y. Lee, and A. Vishwanath, “Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers”, *Phys. Rev. Research* **5**, 023166 (2023). *arXiv:2210.13477*.
15. J. Dong, J. Wang, and L. Fu, “Dirac electron under periodic magnetic field: Platform for fractional Chern insulator and generalized Wigner crystal”, *arXiv:2208.10516*.
14. J. Dong, T. Hartman, and Y. Jiang, “Averaging over moduli in deformed WZW models”, *J. High Energ. Phys.* **2021**, 185 (2021). *arXiv:2105.12594*.
13. J. Dong, V. Juricic, and B. Roy, “Topoelectric circuits: Theory and construction”, *Phys. Rev. Research* **3**, 023056 (2021). *arXiv:2008.11202*.
12. J. Dong and E. Mueller, “Exact Topological Flat Bands from Continuum Landau Levels”, *Phys. Rev. A* **101**, 013629 (2020). *arXiv:1910.08429*.
11. J. Dong, Y. Chen, D. Xu, and Z.-Q. Yin, “Greenberger-Horne-Zeilinger test for multi-dimension and arbitrary time nodes entangled histories”, *Sci. Bull.* **62**(18), pp.1235-1238 (2016). *arXiv:1610.04296*.

Others

10. C. E. Henzinger, J. R. Ehrets, R. Fushio, J. Dong, T. Werkmeister, M. E. Wesson, K. Watanabe, T. Taniguchi, A. Vishwanath, B. I. Halperin, A. Yacoby, and P. Kim, “Controlled localization of anyons in a graphene quantum Hall interferometer”, *arXiv:2603.11182*.
9. A. Valenti, Y. Vituri, Y. Yang, D. E. Parker, T. Soejima, J. Dong, M. A. Morales, A. Vishwanath, E. Berg, and S. Zhang, “Quantum Geometry Driven Crystallization: A Neural-Network Variational Monte Carlo Study”, *arXiv:2512.07947*.
8. M. Uzan, W. Zhi, M. Bocarsly, J. Dong, S. Dutta, N. Auerbach, N. S. Kander, M. Labendik, Y. Myasoedov, M. E. Huber, K. Watanabe, T. Taniguchi, D. E. Parker, and E. Zeldov, “hBN alignment orientation controls moiré strength in rhombohedral graphene”, *arXiv:2507.20647*.
7. T. Soejima, J. Dong, O. E. Sommer, D. E. Parker, and A. Vishwanath, “Topological constraint on crystalline current”, *Phys. Rev. Lett.* **136**, 186601 (2026). *arXiv:2507.18611*.
6. R. Fan, J. Dong, and A. Vishwanath, “Simulating the non-unitary Yang-Lee conformal field theory on the fuzzy sphere”, *arXiv:2505.06342*.

5. P. J. Ledwith, J. Dong, A. Vishwanath, and E. Khalaf, “Nonlocal Moments in the Chern Bands of Twisted Bilayer Graphene”, *Phys. Rev. X* **15**, 021087 (2025). *arXiv:2408.16761*.
4. Q. Li, J. Dong, P. J. Ledwith, and E. Khalaf, “Constraints on real space representations of Chern bands”, *arXiv:2407.02561*.
3. M. Fujimoto, D. E. Parker, J. Dong, E. Khalaf, A. Vishwanath, and P. J. Ledwith, “Higher vortexability: zero field realization of higher Landau levels”, *Phys. Rev. Lett.* **134**, 106502 (2025). *arXiv:2403.00856*. (Editor’s Suggestion, Featured in PRL’s Collection of the Year 2025)
2. Q. Gao, J. Dong, P. J. Ledwith, D. E. Parker, and E. Khalaf, “Untwisting moiré physics: Almost ideal bands and fractional Chern insulators in periodically strained monolayer graphene”, *Phys. Rev. Lett.* **131**, 096401 (2023). *arXiv:2211.00658*. (PRL cover)
1. J. Dong, V. Elser, G. Gyawali, K. Y. Jee, J. Kent-Dobias, A. Mandaiya, M. Renz, and Y. Su, “Glass phenomenology in the hard matrix model”, *J. Stat. Phys.* **2021**(9), 093302 (2021). *arXiv:1912.07558*.

PRESENTATIONS

Invited Talks

- *Electron Crystallography in Rhombohedral Graphene*, Long Ju Group Meeting, MIT, May 2026.
- *Anomalous Hall Crystal and its Phonons*, Kids Seminar, Harvard University, Feb 2026.
- *Anomalous Hall Crystal and its Phonons*, Center for Computational Quantum Physics, Flatiron Institute, Dec 2025.
- *Anomalous Hall Crystal and its Phonons*, Geballe Laboratory for Advanced Materials, Stanford University, Dec 2025.
- *Anomalous Hall Crystal and its Phonons*, Thouless Institute for Quantum Matter Seminar, University of Washington Seattle, Dec 2025.
- *Anomalous Hall Crystal and its Phonons*, MRS Spring Meeting and Exhibit, Apr 2025.
- *Anomalous Hall Crystals in Rhombohedral Multilayer Graphene*, Joonho Lee Group Meeting, Harvard University, Mar 2025.
- *Stripping off the Magnetic Field from the Lowest Landau Level*, KITP Locals Lunch, Aug 2024.
- *Anomalous Hall Crystal in Rhombohedral Multilayer Graphene*, KITP Condensed Matter Meeting, Jul 2024.
- *Anomalous Hall Crystal in Rhombohedral Multilayer Graphene*, Quantum Theory Seminar, Cornell University, Apr 2024.
- *Composite Fermions Form and Flow without a Magnetic Field*, Physical Review Journal Club, American Physical Society, Nov 2023.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted MoTe_2* , Special AEP Seminar, Cornell University, Aug 2023.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted MoTe_2* , Thouless Institute for Quantum Matter Seminar, University of Washington Seattle, Jun 2023.
- *Exact Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers*, Quantum Matter in Mathematics and Physics, Center of Mathematical Sciences and Applications, Harvard University, MA, Dec 2022.

Posters

- *Anomalous Hall Crystals in Rhombohedral Multilayer Graphene*, Thouless Institute for Quantum Matter Winter Workshop, WA, Jan 2024.

- *Composite Fermi Liquid at Zero Magnetic Field in Twisted MoTe₂*, Thouless Institute for Quantum Matter Winter Workshop, WA, Jan 2024.
- *Anomalous Hall Crystals in Rhombohedral Multilayer Graphene*, National High Magnetic Field Laboratory Theory Winter School, FL, Jan 2024.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted MoTe₂*, National High Magnetic Field Laboratory Theory Winter School, FL, Jan 2024.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted MoTe₂*, Princeton Summer School on Condensed Matter Physics 2023, Princeton University, July 2023.
- *Exact Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers*, Spring 2023 meeting of the Simons Collaboration on Ultra-Quantum Matter, CU Boulder, CO, May 2023.
- *Exact Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers*, National High Magnetic Field Laboratory Theory Winter School, FL, Jan 2023.
- *Exact Topological Flat Bands from Continuum Landau Levels*, ARO/AFOSR MURI Program Review Meeting, UMass Amherst, MA, Oct 2019.

TEACHING EXPERIENCE

Teaching Fellow, Fall 2025

Led discussion sections, held office hours, supervised presentations, created and graded homework, and took lecture notes for PHYSICS 268R (Quantum Phases of Matter).

Teaching Fellow, Spring 2025

Led office hours, supervised presentations, and gave lectures for PHYSICS 195B (Introduction to Quantum Materials and Devices).

Teaching Fellow, Fall 2023

Held sections, led office hours, and graded homework and exams for PHYSICS 195A (Introduction to Solid State Physics).

Teaching Assistant, Fall 2019

Held one-hour study halls for PHYS 7651 (Quantum Field Theory 1) every week.

Teaching Assistant, Fall 2020

Graded homework and answers questions online for PHYS 7681 (Quantum Information Processing).

SERVICE

APS March Meeting Session Chair, 2025

Chaired session J27: Emergent Phases and Transport Phenomena in Twisted and Multilayer Graphene Systems.

APS March Meeting Session Chair, 2024

Chaired session Z07: Magnetic Topological Semimetals III.

Journal Referee, 2023-Present

Provided peer review for:

- Nature (×2)
- Physical Review X (×8)
- Physical Review Letters (×6)
- Physical Review Research (×3)
- Physical Review B (×30)
- Journal of Physics: Condensed Matter (×3)
- Reports of Progress in Physics: Original Research (×1)

Cornell Alumni Admissions Ambassadors Network Volunteer, 2024

Met with prospective applicants to discuss details about undergraduate experience at Cornell.

Harvard Organ Society Recital Coordinator, 2022-2024

Revived the Busch Midday Recital Series after the COVID pandemic. Invited and hosted professional organists. Doubled audience size during tenure.